
Reassessing Ancient Connectivity in Central Italy from Fragmentary Sources: A Multiscale Uncertainty-Aware Network Model

Caterina Paola Venditti^{*1,2}

¹University of Cassino and Southern Lazio — European University of Technology (UNICAS EUT+) – Italy

²Ministry of Culture (Italy) - Superintendency for Archaeology, Fine Arts and Landscape for the Provinces of Frosinone and Latina (MiC — SABAP FR-LT) – Italy

Abstract

Long-term topographical research has highlighted the strategic role of the landscapes of Central Italy, where processes of mobility, frontier interaction, regional integration, and local connectivity unfolded across multiple spatial and chronological scales.

Focusing on Southern Latium and its bordering regions, including northern Campania and the Apennine areas of Molise and Abruzzo between the 2nd century BC and Late Antiquity, complex patterns of connection emerge around key mobility corridors, peripheral landscapes, and local settlement systems. Formal network modelling provides the analytical framework to structure and evaluate these relationships systematically, making it possible to address three interconnected questions: (1) whether marginal settlements were integrated into broader regional systems; (2) whether border areas gravitated towards distinct networks compared to major corridors; and (3) whether minor sites developed autonomous dynamics at the micro-scale.

Formalising these interactions requires a multilayer analytical approach capable of evaluating degrees of integration, gravitational behaviours, and clustering tendencies. Building such a model means confronting the heterogeneous data (spanning surveys, excavations, cartography and legacy documentation), which vary in spatial density, chronological depth, and reliability. Major urban centres, sacred places, imperial estates, monumental villas and Late Antique episcopal seats are generally well documented, whereas knowledge of rural micro-areas is much more fragmentary due to the uneven intensity of past research. A similar disparity characterises the road system: the main NW–SE corridors (Via Severiana, Via Appia, Via Latina, and the Corfinium–Sepinum route), together with the principal transversal routes linking the Tyrrhenian and Adriatic sectors, including the Via Valeria, are relatively well attested, while E–W connectors and routes at the scale of individual agri remain poorly documented. As many of these datasets were produced for purposes other than formal network analysis, their integration results in a patchy spatial coverage and variable resolution, posing significant challenges for structural analysis.

To address these issues, a multiscale, source-sensitive approach is adopted, integrating archaeological and topographical evidence with landscape analysis within a GIS project. Connectivity is expressed through weighted, predominantly undirected edges representing documented or inferred routes, with computational inference reconstructing missing segments.

^{*}Speaker

The graph structure integrates hierarchically differentiated nodes, including urban centres, sacred places, harbours, monumental rural complexes, imperial villas, mobility nodes, villages, and rural sites. Geo-landscape features such as valleys, passes, river crossings, and slope, are incorporated as spatial constraints and weighting factors influencing connectivity patterns and route reconstruction. Uncertainty is treated as an explicit analytical dimension and encoded within the geodatabase through chronological ranges, confidence scores, and reliability levels. Specifically, these attribute values function as a topological filter: segments exceeding a defined uncertainty threshold are dynamically excluded or penalized in the graph structure. This allows for multi-scenario connectivity testing, isolating the highly reliable core network from more speculative connections. A multi-resolution strategy is adopted to connect major nodes at the regional scale, introducing micro-level links only where the evidence can plausibly support them. This bottom-up approach ensures that network structure emerges from the sources themselves rather than from *a priori* assumptions. The framework also compares different levels of network, ranging from theoretical proximity-based models to geographically constrained networks shaped by landscape elements and historically documented infrastructural systems, evaluating how different premises affect network topology and centrality patterns.

Progressively enriched scenarios explore shifts in relational configurations as different node classes are incorporated, highlighting the methodological consequences of converting discontinuous evidence. Comparative analysis reveals that different reconstruction strategies produce substantially different centrality patterns, demonstrating how interpretative choices influence historical reconstructions. Explicit uncertainty representation significantly affects the identification of hubs, peripheral areas, and mobility corridors, while multiscale analysis shows that marginal landscapes were not uniformly peripheral but participated in regional systems through scale-dependent forms of connectivity.

Preliminary results demonstrate how uncertainty-aware and multiscale approaches transform network analysis into a tool for interrogating, rather than simply representing, incomplete archaeological datasets, underscoring the heuristic value of formal modelling for reassessing connectivity.

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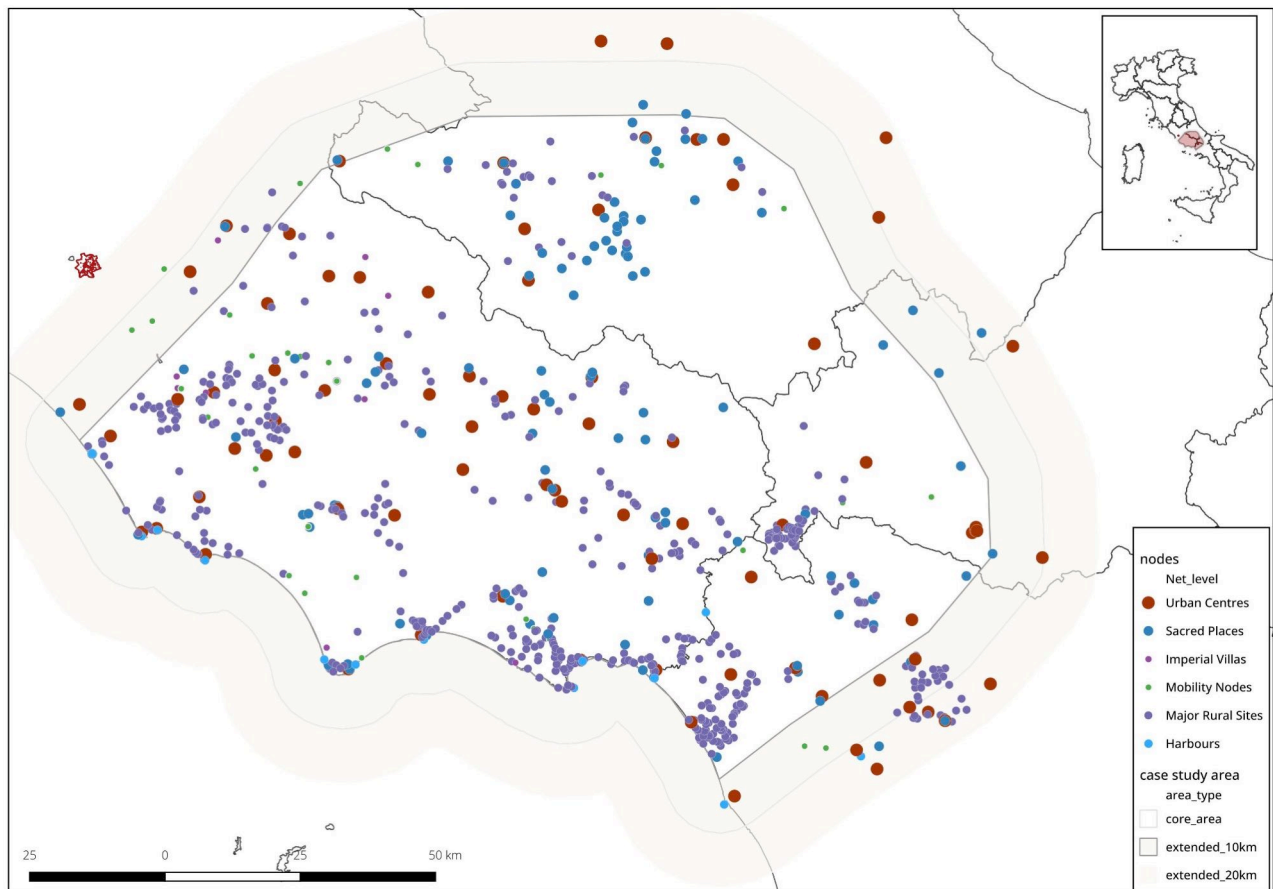


Figure 1. Archaeological nodes incorporated into the network models, including urban centres, sacred places, monumental rural sites, imperial villas, mobility nodes, and harbours. The study area is structured into a core zone and two surrounding buffer zones (10 km and 20 km) to reduce edge effects during network analysis. Source: Author's work.

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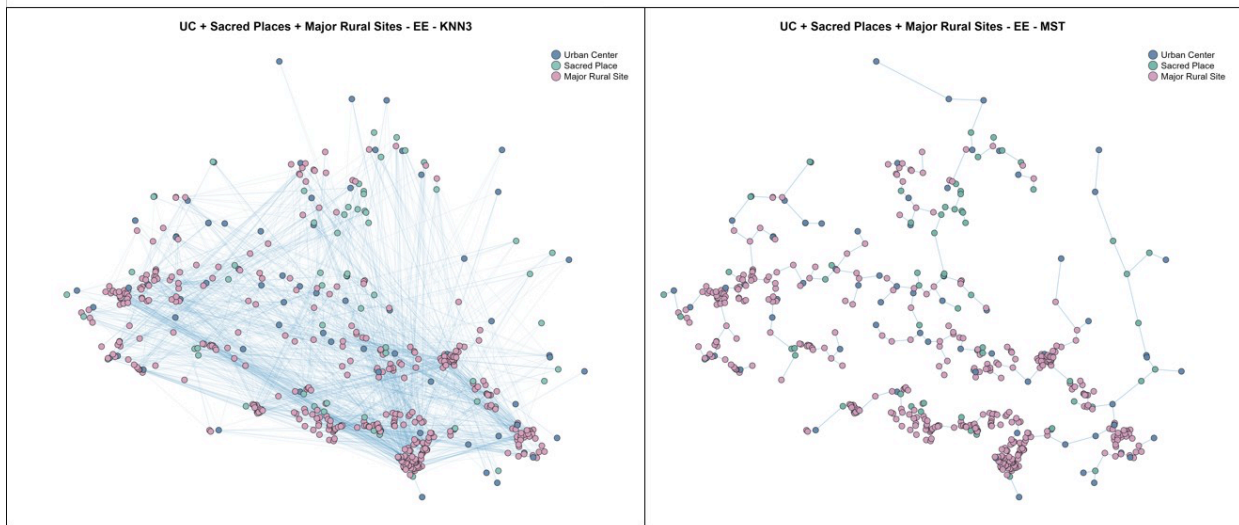


Figure 2. Comparison of two theoretical network models generated from the same archaeological dataset (Urban Centres, Sacred Places, and Major Rural Sites) for the Early Empire phase. (A) K-nearest neighbours model (KNN3), in which each node is connected to its three nearest neighbours, producing a dense proximity-based network. (B) Minimum Spanning Tree (MST), which connects all nodes through the minimum number of links required to maintain network connectivity. The comparison illustrates how different network construction rules generate markedly different topological structures from the same archaeological evidence. Source: Author's work.

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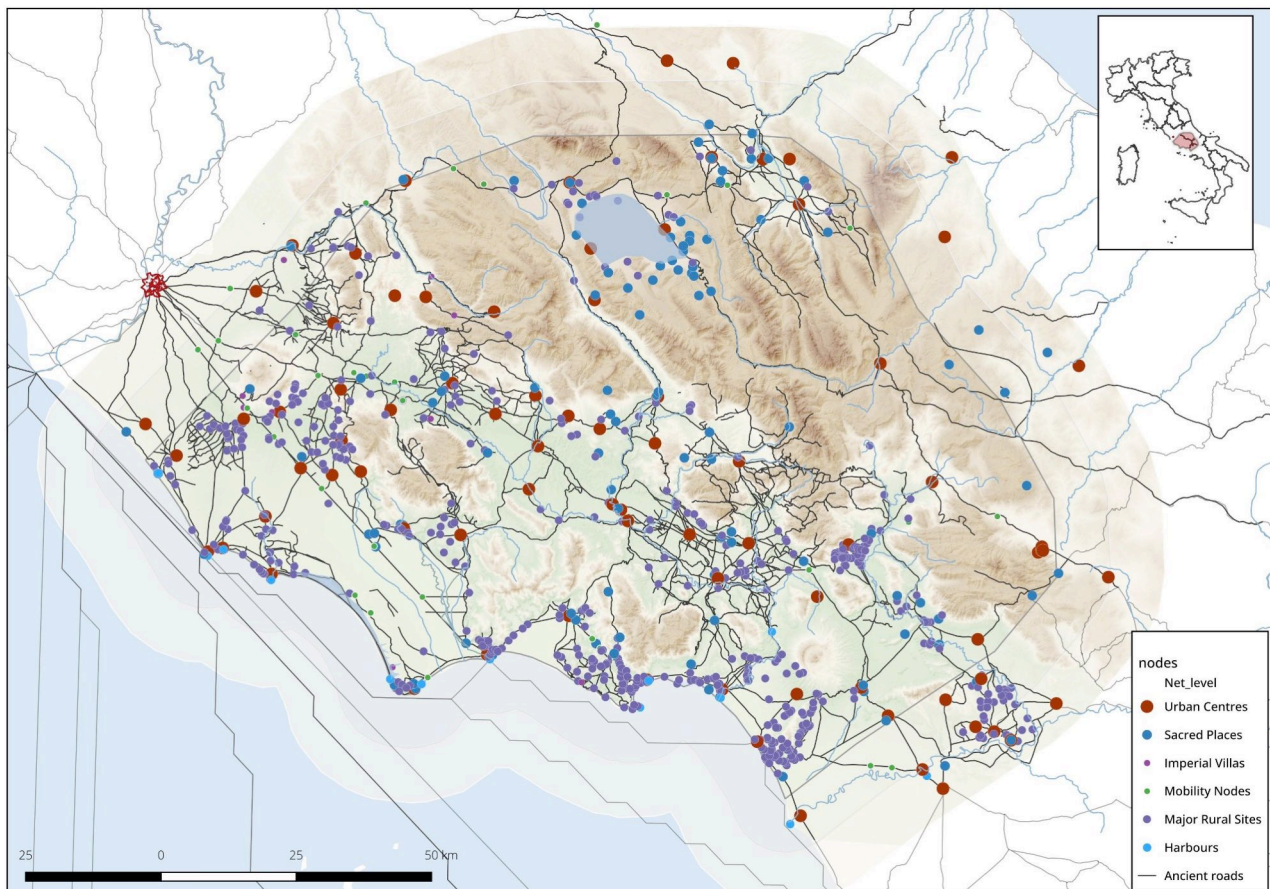


Figure 3. Preliminary integration of archaeological nodes, reconstructed road infrastructure, and topographic context. This figure represents an intermediate stage in the development of the geographically constrained network model and is intended to illustrate the ongoing construction of the analytical framework. Source: Author's work.